

KALQUEST[®] Powder for Oral Suspension 5 g/sachet

Calcium Polystyrene Sulfonate 5.0 g

Product Description

Before reconstitution: fine powder, cream colour, mocha aroma, and slightly sweet

After reconstitution: suspension, cream colour, mocha aroma, and slightly sweet

Pharmacodynamics

Kalquest[®] is a cation exchange resin which releases calcium in the intestines and binds potassium. This reduces the absorption and metabolic availability of potassium.

Pharmacokinetics

Not applicable as this product is not absorbed.

Indication

Kalquest[®] is used to treat hyperkalaemia resulting from acute or chronic renal failure.

Recommended Dosage

15 - 30g daily in 2 -3 divided doses. Each dose should be suspended in 30 - 50ml of water and administered orally.

The dosage recommendations are a guide only; the precise requirements should be decided on the basis of regular serum electrolyte determinations.

Route of Administration

Oral

Contraindications

- In patients with plasma potassium levels below 5 mmol/litre
- Conditions associated with hypercalcaemia (e.g. hyperparathyroidism, multiple myeloma, sarcoidosis or metastatic carcinoma)
- History of hypersensitivity to polystyrene sulfonate resins
- Obstructive bowel disease
- Kalquest[®] should not be administered orally to neonate
- Hypersensitivity to the active substance or to any of the excipients

Warning and Precautions

Sorbitol: Gastrointestinal stenosis, intestinal ischemia and its complications (necrosis and perforation) may occur in patients treated with polystyrene sulfonate, especially in patients using sorbitol. Therefore concomitant use of Sorbitol with calcium polystyrene sulfonate is not recommended.

Hypokalaemia: The possibility of severe potassium depletion should be considered and adequate clinical and biochemical control is essential during treatment, especially in patients on digitalis. Administration of the resin should be stopped when the serum potassium falls to 5 mmol/litre.

Other electrolyte disturbances: Like all cation-exchange resins, calcium polystyrene sulfonate is not totally selective for potassium. Hypomagnesaemia and/or hypercalcaemia may occur. Accordingly, patients should be monitored for all applicable electrolyte disturbances. Serum

calcium levels should be estimated at weekly intervals to detect the early development of hypercalcaemia, and the dose of resin adjusted to levels at which hypercalcaemia and hypokalaemia are prevented.

Other risks: In the event of clinically significant constipation, treatment should be discontinued until normal bowel movement has resumed. Magnesium containing laxatives should not be used. The patient should be positioned carefully when ingesting the resin, to avoid aspiration, which may lead to bronchopulmonary complications.

Children and neonates: In neonates, calcium polystyrene sulfonate should not be given by the oral route. Due to the risk of digestive haemorrhage or colonic necrosis, particular care should be observed in premature infants or low birth weight infants.

Interaction with Other Medicaments

Concomitant use not recommended

Sorbitol (oral or rectal): Concomitant use of Sorbitol with calcium polystyrene sulfonate is not recommended due to cases of intestinal necrosis and other serious gastrointestinal adverse reactions, which may be fatal.

To be used with caution

Cation-donating agents: may reduce the potassium binding effectiveness of Kalquest®.

Non-absorbable cation-donating antacids and laxatives: There have been reports of systemic alkalosis following concurrent administration of cation exchange resins and non-absorbable cation-donating antacids and laxatives such as magnesium hydroxide and aluminium carbonate.

Aluminium hydroxide: Intestinal obstruction due to concretions of aluminium hydroxide has been reported when aluminium hydroxide has been combined with the resin (sodium form).

Digitalis-like drugs: The toxic effects of digitalis on the heart, especially various ventricular arrhythmias and A-V nodal dissociation, are likely to be exaggerated if hypokalaemia and/or hypercalcaemia are allowed to develop

Lithium: Possible decrease of lithium absorption.

Levothyroxine: Possible decrease of levothyroxine absorption.

Pregnancy and Lactation

No data are available regarding the use of polystyrene sulfonate resins in pregnancy and lactation. The administration of Kalquest® in pregnancy and during breastfeeding therefore is not advised unless, in the opinion of the physician, the potential benefits outweigh any potential risks.

Side Effects

- **Metabolism and nutrition disorders**

In accordance with its pharmacological actions, the resin may give rise to hypokalaemia and hypercalcaemia, and their related clinical manifestations.

Cases of hypomagnesaemia have been reported.

Hypercalcaemia has been reported in well dialysed patients receiving calcium resin, and occasionally in patients with chronic renal failure. Many patients in chronic renal failure

have low serum calcium and high serum phosphate, but some, who cannot be screened out beforehand, show a sudden rise in serum calcium to high levels after therapy with calcium resin. The risk emphasises the need for adequate biochemical control.

- **Gastrointestinal disorders**

Gastric irritation, anorexia, nausea, vomiting, constipation and occasionally diarrhoea may occur. Gastrointestinal concretions (bezoars) following oral administration have been reported. Gastrointestinal stenosis and intestinal obstruction have also been reported, possibly, due to co-existing pathology or inadequate dilution of the resin.

Gastrointestinal ischemia, ischemic colitis, gastro-intestinal tract ulceration or necrosis, which could lead to intestinal perforation have been reported which is sometimes fatal. The majority of cases have been reported with concomitant use of Sorbitol.

- **Respiratory, thoracic and mediastinal disorders**

Some cases of acute bronchitis and/or broncho-pneumonia associated with inhalation of particles of calcium polystyrene sulfonate have been described.

Symptoms of Overdose and Treatment

Biochemical disturbances from over dosage may give rise to clinical signs or symptoms of hypokalaemia, including irritability, confusion, delayed thought processes, muscle weakness, hyporeflexia and eventual paralysis. Apnoea may be a serious consequence of this progression. Electrocardiographic changes may be consistent with hypokalaemia or hypercalcaemia; cardiac arrhythmia may occur. Appropriate measures should be taken to correct serum electrolytes and the resin should be removed from the alimentary tract by appropriate use of laxatives or enemas.

Storage Condition

Store at below 30°C

Dosage Form and Packaging

Box contains of 30 sachets @ 5 g oral powder for suspension

Shelf Life

24 months

Registration Number

KALQUEST® Powder for Oral Suspension 5g/sachet - MALXXXXXXXX

Name and Address of Manufacturer

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Product Registration Holder

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